EVERY OBJECT OF THE PRO-ETALE SITE HAS A HYPERCOVERING WHOSE TERMS ARE SINGLE WEAKLY CONTRACTIBLE OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the application of the hypercovering construction to finite coverings by weakly contractible objects, the passage to degreewise disjoint unions, and the repetition of the argument inside the subcategory of quasi-compact and quasi-separated objects with affine weakly contractible coverings.

Editorial note. The label lemma-w-contractible-hypercovering names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-w-contractible-hypercovering –start 3215 –end 3266 –local-id 3579 returned RESULT=OK (L3579 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Coverings by weakly contractible objects. Same chapter, the lemma that every object of the pro-etale site has a finite covering by weakly contractible objects, and the lemma producing coverings by affine weakly contractible objects for quasi-compact and quasi-separated objects. Supply the input coverings to which the hypercovering construction is applied.
- Construction of hypercoverings from coverings with bounded index sets. Hypercoverings, the lemma producing a hypercovering with terms indexed by finite sets from coverings whose index sets are bounded by a cardinal. Supplies the hypercoverings K with finitely many weakly contractible terms in each degree.
- Replacing a hypercovering by its degreewise disjoint unions. Hypercoverings, the remark that a hypercovering may be replaced by the hypercovering whose degree n term is the disjoint union of the terms in degree n. Supplies the passage to hypercoverings whose every term is a single object.

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- Continuity of the inclusion of the quasi-compact quasi-separated objects. Hypercoverings, the lemma that a continuous functor commuting with fibre products carries hypercoverings to hypercoverings. Supplies the passage from hypercoverings in the subcategory of quasi-compact and quasi-separated objects to hypercoverings in the pro-etale site.

Source reference. Original lemma 0DAZ: sources/stacks-chow/tags.txt, source lines 16679–16679. Prerequisite pointer invoked at proof line(s) 3252; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 094K: sources/stacks-chow/tags.txt, source lines 11553–11553. Prerequisite pointer invoked at proof line(s) 3234; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0DB2: sources/stacks-chow/tags.txt, source lines 16682–16682. Prerequisite pointer invoked at proof line(s) 3240; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 098I: sources/stacks-proetale/pinned-original-source.tex, source lines 3087–3087. Prerequisite pointer invoked at proof line(s) 3255; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0F4P: sources/stacks-proetale/pinned-original-source.tex, source lines 3143–3143. Prerequisite pointer invoked at proof line(s) 3233; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme.*

(1) *For every object U of $X_{pro\text{-}étale}$ there exists a hypercovering K of U in $X_{pro\text{-}étale}$ such that each term K_n consists of a single weakly contractible object of $X_{pro\text{-}étale}$ covering U.*
(2) *For every quasi-compact and quasi-separated object U of $X_{pro\text{-}étale}$ there exists a hypercovering K of U in $X_{pro\text{-}étale}$ such that each term K_n consists of a single affine and weakly contractible object of $X_{pro\text{-}étale}$ covering U.*

Proof. Let $\mathcal{B} \subset \mathrm{Ob}(X_{pro\text{-}étale})$ be the set of weakly contractible objects of $X_{pro\text{-}étale}$. Every object T of $X_{pro\text{-}étale}$ has a covering $\{T_i \to T\}_{i\in I}$ with I finite and $T_i \in \mathcal{B}$ by Lemma 0F4P. By Hypercoverings, Lemma 094K we get a hypercovering K of U such that $K_n = \{U_{n,i}\}_{i\in I_n}$ with I_n finite and $U_{n,i}$ weakly contractible. Then we can replace K by the hypercovering of U given by $\{U_n\}$ in degree n where $U_n = \coprod_{i\in I_n} U_{n,i}$ This is allowed by Hypercoverings, Remark 0DB2.

Let $X_{qcqs,pro\text{-}étale} \subset X_{pro\text{-}étale}$ be the full subcategory consisting of quasi-compact and quasi-separated objects. A covering of $X_{qcqs,pro\text{-}étale}$ will be a finite pro-étale covering. Then $X_{qcqs,pro\text{-}étale}$ is a site, has fibre products, and the inclusion functor $X_{qcqs,pro\text{-}étale} \to X_{pro\text{-}étale}$ is continuous and commutes with fibre products. In particular, if K is a hypercovering of an object U in $X_{qcqs,pro\text{-}étale}$ then K is a hypercovering of U in $X_{pro\text{-}étale}$ by Hypercoverings, Lemma 0DAZ. Let $\mathcal{B} \subset \mathrm{Ob}(X_{qcqs,pro\text{-}étale})$ be the set of affine and weakly contractible objects. By Lemma 098I and the fact that finite unions of affines are affine, for every object U of $X_{qcqs,pro\text{-}étale}$ there exists a covering $\{V \to U\}$ of $X_{qcqs,pro\text{-}étale}$ with

$V \in \mathcal{B}$. By Hypercoverings, Lemma 094K we get a hypercovering K of U such that $K_n = \{U_{n,i}\}_{i \in I_n}$ with I_n finite and $U_{n,i}$ affine and weakly contractible. Then we can replace K by the hypercovering of U given by $\{U_n\}$ in degree n where $U_n = \coprod_{i \in I_n} U_{n,i}$. This is allowed by Hypercoverings, Remark 0DB2. □